CSE428

Image Processing

Assignment 1

Compare and Contrast State-of-the-Art CNN Architectures Used in Image Processing with the Vision Transformer

Name: Muaz Ibne Ashraf
Student ID: 21101177
Section: 01

Date: 10 May 2026

Abstract

This report illustrates a comparison among the following architectures: VGG-16, GoogLeNet/Inception v1, ResNet-50, DenseNet-121, EfficientNet-B0, and Vision Transformer ViT-B/16, which are significant in the context of image processing. The comparison highlights that CNN progress was not solely based on increasing the depth of CNN models, but also involved the effective use of computation through the use of small kernels, multi-branch modules, residual shortcuts, dense feature reuse, and balanced scaling. However, there exists a class of designs that represent a different philosophy of design: ViT-B/16 is one such design, treating an image as a series of patches, and using self-attention to model global relationships. The main conclusion is that CNNs remain strong when local spatial bias, data efficiency, and practical deployment are important, while ViT becomes most attractive when large-scale pretraining and global context modeling are available.

1 Introduction

Deep learning architectures that learn the visual features directly from the data have revolutionized image processing. The design of features has been crucial for many computer vision systems in the past and modern neural networks learn hierarchical representations from the pixels. The six architectures in this report all had to address the same primary question: how to increase the accuracy of the model while ensuring a manageable amount of compute power for the model and keeping the model appropriate for tasks in the real world.

CNN models in this comparison share a fundamental principle: they employ convolutional filters to make use of the locality in an image. This induces a bias towards visual information in CNNs since neighbouring pixels are likely to be correlated and similar features are likely to be found at various points in the image. These are the principles that are extended by VGG-16, GoogLeNet, ResNet-50, DenseNet-121, and EfficientNet-B0. In the other way, ViT-B/16 replaces the convolution and uses the Transformer concept with image patches. This reduces the reliance on handcrafted visual structure, increases reliance on large datasets, and also increases pretraining reliance in ViT.

2 Architectural Design and Key Innovations

VGG-16 is one of the most prominent works related to a deep and uniform CNN. It lastly applies repeated 3×3 convolution filters, max-pooling layers and fully connected layers. Its one of the key features was the demonstration that many small filters could be stacked together to produce a big effective receptive field, and more nonlinear transformations. This design was simple and straightforward and very convenient to use as a base model, but also had a huge number of parameters, around 138 million, largely due to the fully connected layers [1].

The change of the design direction from simple depth to multi-scale processing happened with GoogLeNet, also called Inception v1. It runs 1 x 1, 3 x 3, 5 x 5 and pooling operations in parallel and then concatenates them. The 1 x 1 convolutions decrease the number of channels prior to the expensive operations, allowing the model to be deep and wide, while having a reasonable computational budget. GoogLeNet is 22 layers deep, and designed to get better use of computation within the network [2].

ResNet-50 addressed a key issue in training very deep networks. However, as CNNs got deeper, plain networks could experience degradation, meaning that adding more layers would make optimization more difficult than it would be beneficial. ResNet proposes to add a residual mapping to an identity

shortcut in the form of residual blocks. This makes it easier to implement gradients and also makes deep networks viable. ResNet-50 is an efficient model with high accuracy and is widely applicable across various fields, including classification, detection, and segmentation, as a backbone model [3].

DenseNet-121 takes the concept of connectivity further. The DenseNet concatenates feature maps of all its previous layers instead of adding them earlier via an identity shortcut. This promotes the reuse of features, promotes robust feature propagation, and mitigates redundant learning. DenseNet-121 is considerably smaller than some of the previous CNNs; however, in training, the densely connected architecture can lead to memory consumption due to the high number of intermediate feature maps that have to be stored [4].

EfficientNet-B0 is concentrating on efficiency and scaling. Instead of scaling a model just in depth, just in width or just in input resolution, EfficientNet suggests compound scaling, which enlarges all three parameters in a balanced manner. B0 is a baseline model that is built using a neural architecture search framework and the MBConv blocks (with squeeze and excitation). This renders EfficientNet-B0 particularly appealing when both precision and number of parameters, FLOPs and deployment are important [5].

ViT-B/16 is structurally different from the CNNs. It breaks down an image into 16x16 patches, maps each patch to an embedding, appends positional information, and applies a sequence of a Transformer encoder. The main contribution is demonstrating the feasibility of using a pure Transformer without convolution as the primary operation in image recognition tasks. ViT has, however, a lower intrinsic locality and translation bias compared to CNNs, and requires larger-scale pretraining to achieve the best performance [6].

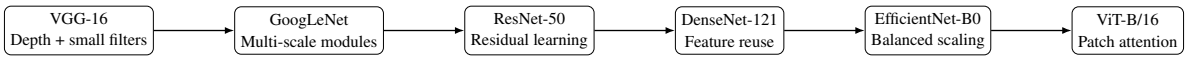


Figure 1: Evolution of the dominant design idea across the selected architectures.

3 Computational Complexity and Performance Trends

The performance improvement from VGG-16 to ViT-B/16 is not straightforwardly a ranking as the original papers applied various training settings, test-time procedures, and pretraining datasets. Because of this, this report is not a strict leaderboard, but rather represents numbers as a representative paper-reported result. When a paper reports ensemble, crop-based, or post submission architectural behavior, these are only reported as context, unless there is a single model architectural behavior reported and the ensemble, crop-based, or post submission behavior is not entirely different. The most straightforward comparison is architectural: VGG achieved performance gains by stacking layers; GoogLeNet achieved performance gains from multi-scale modules and dimension reduction; ResNet achieved performance gains from having easier-to-optimize network; DenseNet achieved performance gains through reusing features; EfficientNet achieved performance gains through balancing scaling; and ViT achieved performance gains by using global self-attention and large-scale pretraining.

Table 1: Representative Architectural and Efficiency Comparison

Model	Core Design	Scale / Complexity Signal	Representative Performance Trend	Citation
VGG-16	Sequential CNN using repeated 3×3 convolutions and FC layers	Very high parameters, about 138M	Strong ImageNet-era baseline, but computationally heavy	[1]
GoogLeNet	Inception modules with parallel multi-scale branches	22 layers; dimension reduction with 1×1 convolutions	High accuracy with better compute use than plain deep CNNs	[2]
ResNet-50	Residual bottleneck blocks and identity shortcuts	About 25.6M parameters; 3.8B FLOPs in paper setting	Strong general-purpose accuracy and excellent trainability	[3]
DenseNet-121	Dense blocks with feature-map concatenation	Compact parameter use, but activation memory can be high	Competitive accuracy through feature reuse and deep supervision	[4]
EfficientNet-B0	MBConv baseline with compound scaling principle	About 5.3M parameters and 0.39B FLOPs	Best efficiency-oriented CNN among the selected models	[5]
ViT-B/16	16×16 patch embeddings and Transformer encoder	About 86M parameters; pretraining scale is critical	Very strong with large-scale pretraining and transfer learning	[6]

While the depth is useful for better feature learning, VGG-16 also demonstrates the price of merely stacking layers and applying huge fully connected classifiers. To address that weakness, GoogLeNet makes an improvement in compute utilization within itself. ResNet-50 then switches the primary consideration from module width to optimization stability, which makes deep networks easier to train. Additionally, DenseNet-121 enhances information flow through direct access to previous features, providing a more seamless user experience. EfficientNet-B0 is a more recent CNN paradigm: assess the model based on accuracy per parameter and accuracy per FLOP, rather than the top-line accuracy.

ViT-B/16 should be brought up with a fairness note. It's fine and it's helpful to acknowledge that ViT relies on a larger pretraining dataset, since that is a key architectural constraint. But it does not imply that ViT-B/16 can be treated like older CNNs and trained under the same conditions. With the latter, the conclusion is that CNNs are more data efficient due to Their local vision bias, and ViT can be extremely competitive if the scale is sufficient for the model to learn the visual structure from data [6].

4 Suitability for Image Processing Tasks

If classification of images is the target, typically ResNet-50 is the most useful existing base network for general purpose classification, as it is reliable for training, has manageable computational cost and has good transfer ability. EfficientNet-B0 is better suited for systems with stringent memory or latency requirements or for mobile devices. DenseNet-121 is acceptable for scenarios where memory consumption during training needs to be taken into account but is useful for situations where compact parameterization and feature reuse are essential.

ResNet-style backbones are more practical than VGG for object detection and segmentation tasks due to the ability of residual blocks in deeper feature extractors and stable optimization. The idea of GoogLeNet's inception is helpful for multi-scale visual patterns; however, the module design is more complicated than ResNet. Although VGG is still found valuable for training, feature extraction, and research comparison, it is not ideal for deployment on modern low-resource systems due to a high number of parameters.

ViT-B/16 is particularly well suited to tasks in which the global context is important, and in which there is strong pre-training. It can directly model the long range relationship with a self-attention mech-

anism, which is more direct than the normal local convolution. However, ViT is not always the optimal solution for image processing applications. For smaller datasets or hardware that doesn't have much power, a CNN like ResNet-50 or EfficientNet-B0 is a better bet to be reliable and easier to train. In addition to that, hybrid CNN-Transformer-based systems are also appealing due to their hybrid nature of local visual bias and global attention.

Table 2: Strengths, Limitations, and Practical Suitability

Architecture	Main Strengths	Main Limitations	Most Suitable Use Cases
VGG-16	Simple, regular, strong feature hierarchy	Large parameter count and expensive inference	Educational baseline; transfer-feature extraction
GoogLeNet	Multi-scale representation with controlled computation	Complex module design; some reported results use ensembles/crops	Classification and localization with multi-scale patterns
ResNet-50	Stable training and strong transfer performance	Less efficient than newer compact CNNs	General-purpose classification, detection, and segmentation backbone
DenseNet-121	Feature reuse and compact parameters	Activation memory and implementation complexity	Parameter-conscious recognition tasks
EfficientNet-B0	Excellent accuracy-efficiency trade-off	Architecture is less transparent than VGG/ResNet	Mobile and low-latency image classification
ViT-B/16	Global context modeling and strong transfer at scale	Requires large pretraining data and compute to shine	Large-scale transfer learning; context-heavy recognition tasks

5 CNNs vs Vision Transformer

The key distinction between CNNs and ViT lies in the way they perceive the visual information.The main difference between CNNs and ViT is where the visual understanding comes from. CNNs learn image knowledge using local filters, shared weights and progressive spatial abstraction. These assumptions enable CNNs to be efficient and reliable when the data is limited. VGG, GoogLeNet, ResNet, DenseNet, and EfficientNet all embody this key concept, with novel advancements on depth, connectivity, efficiency, and scaling. The drawback of their weakness is that most long-range relationships only model many local operations.

The assumption of ViT is different. It supports self-attention to learn relationships between any two image patches and uses image patches as tokens. This provides a natural approach to modelling global context for ViT, but may also help to mitigate inbuilt image-specific bias. Thus, the effectiveness of ViT relies heavily on the amount and quality of the pretraining data. Not a bug to be cloaked – one of the big lessons of architecture. When reporting, it should be presented as an alternative to CNNs on a scale basis and not necessarily as a replacement for all CNNs

6 Conclusion and Recommendations

It is clearly observed from the six architectures that there is a historical progression in image processing. VGG-16 demonstrated the importance of deep networks and small convolution filters. GoogLeNet demonstrated that multi-scale branching can be used to enhance the utilization of computation. The identity shortcuts enabled ResNet-50 to make deep CNNs practical. DenseNet-121 took feature reuse with dense connectivity to the next level. EfficientNet-B0 demonstrated that scaling up depth, width and resolution together leads to an improvement in accuracy and efficiency. Pre-trained scale is large enough, ViT-B/16 demonstrated the feasibility of self-attention over image patches to beat CNNs.

There is no universal architecture that is best for all image processing applications. If you want to go with a balanced baseline, ResNet-50 is the best. Among the presented CNNs, EfficientNet-B0 is the best one for efficient deployment. DenseNet-121 is the most unique one for feature reuse study. VGG-16 and GoogLeNet are still key to understanding the development of CNN in history. ViT-B/16 is the latest of the models, and works best for large-scale transfer learning and global context modeling. So the choice of practical in the end should depend on the size of the dataset, the compute budget, target task and if there is a large scale pretraining.

The following practical decision guide can be formulated: The first is to select ResNet-50 for a project that requires a robust baseline that is easily defensible from an academic perspective. Secondly, for deployment cost, mobile inference or low latency, select EfficientNet-B0. Thirdly, if compact feature reuse and powerful gradient flow is desired, DenseNet-121 should be chosen. Fourth, select ViT-B/16 when the pretrained weights of the Transformer can be used and when long-range visual relationships or global ones are beneficial. These recommendations relate directly to the performance of the models in architecture and computation, and their applicability to various image-processing tasks.

References

[1] K. Simonyan and A. Zisserman, "Very Deep Convolutional Networks for Large-Scale Image Recognition," in *International Conference on Learning Representations (ICLR)*, 2015.

[2] C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. Reed, D. Anguelov, D. Erhan, V. Vanhoucke, and A. Rabinovich, "Going Deeper with Convolutions," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2015, pp. 1–9.

[3] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.

[4] G. Huang, Z. Liu, L. van der Maaten, and K. Q. Weinberger, "Densely Connected Convolutional Networks," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 4700–4708.

[5] M. Tan and Q. V. Le, "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks," in *Proceedings of the 36th International Conference on Machine Learning (ICML)*, 2019, pp. 6105–6114.

[6] A. Dosovitskiy *et al.*, "An Image is Worth 16 $\times$ 16 Words: Transformers for Image Recognition at Scale," in *International Conference on Learning Representations (ICLR)*, 2021.